
ENGINEER 3PX3

Engineering Economics

Course Outline – Winter Semester 2026



1. Course Details

Lecture Instructional Team:

Dr. Vince Leung, PhD, P.Eng.

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Instructional Assistant Intern:

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Support Staff:

Mr. Basem Yassa

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Mr. Brendan Fallon

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Office Hours: Available starting Week 2 to assist you with course material and address any questions you may have. Please note that the final week for office hours will be Week 12.

Bianka: Monday 2pm-3pm

Imaad: Tuesday 12:30pm-1:30pm

1. Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)

Top Hat Lecture Sections:

- [C01](#)
- [C02](#)
- [C03](#)

2. Communication Policy

Need help and have questions related to the course? We want to make sure your questions get answered. To ensure this, it's important that the correct communication method is used.

Got a question during scheduled class hours (e.g., lecture, Design Studio)? The best way to get a prompt response is to raise your hand and ask away! It is likely that others have the same question too and we can't answer a question that is not asked.

Got a question outside scheduled class hours? In this case, email is the best method. Please direct all emails to: prof3px3@mcmaster.ca. This ensures your email gets directed to the most appropriate individual for the fastest response. Every attempt will be made to reply within 24 hours (excluding weekends). Please include a subject prefix of "ENGINEER 3PX3". Emails must be sent from your @mcmaster.ca account. Be sure to include your student number in your email.

3. Inclusive Environment Statement

We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability – and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming, and inclusive environment for every other member of the class.

We will gladly honour your request to address you by an alternate name or gender pronoun. Please advise us of this preference early in the semester so that we may make appropriate changes to our records.

4. Course Description and Learning Outcomes

Engineering economic analysis is explored in the context of an individual engineering project and an interdisciplinary engineering project. Students perform cost-benefit analysis considering the perspective of the individual and engineering team, client, and society in order to maximize the net value of an engineering decision. Factors considered and incorporated include conversion factors, opportunity cost, time value of money, risk analysis, and technical analysis.

LEARNING OUTCOMES

Upon successful completion of the course, the student should be able to:

LO.01	Utilize economic principles to make decisions in engineering projects
LO.02	Formulate Net Value functions to evaluate and compare the value and cost of alternative engineering decisions
LO.03	Make reasonable assumptions or perform necessary research to cope with ambiguity and uncertainty in required tasks
LO.04	Apply the fundamentals of cost, price, present value, and other financial metrics
LO.05	Manage group projects and interpersonal relations, especially building project management skills, preparing agendas, and diagnosing team dynamics
LO.06	Professionally communicate complex engineering decisions following an economic analysis

The Graduate Attributes defined in this section are measured for Accreditation purposes only and will not be directly taken into consideration in determining a student's actual grade in the course. For more information on Accreditation, please visit: <https://www.engineerscanada.ca>. Mapping of the Learning Outcomes outlined above to the Canadian Engineering Accreditation Board (CEAB) Graduate Attributes are outlined in the table below:

MAPPING TO GRADUATE ATTRIBUTES

GRADUATE ATTRIBUTE	LEARNING OUTCOMES
A04 Design	
4.1 Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions.	LO1, LO2
4.2 Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements.	LO1, LO2
4.3 Develops models or prototypes, tests, evaluates, and iterates as appropriate.	LO1, LO2
A06 Individual and Teamwork	
6.1 Actively contributes to the planning and execution of a team project.	LO5
A07 Communication Skills	
7.1 Demonstrates comprehension of technical and non-technical instructions and questions.	LO6
A11 Economics and Project Management	
11.1 Applies economic principles in decision making.	LO1, LO2, LO4
11.2 Plans and effectively manages a project's time, resources, and scope, following business practices as appropriate.	LO3, LO5
11.3 Identifies, characterizes, assesses, and manages risks to project success.	LO3

5. Course Delivery

The delivery of this course occurs through weekly lectures and Design Studios. The course consists of three sections: C01, C02, and C03. You may attend any lecture section, but if you want the attendance grade, you must log your attendance with TopHat appropriately.

The course is run through two **50-minute** lectures and one **1-hour 50-minute** Design Studios weekly.

Lecture topics change each week and support the succeeding Design Studios. Refer to your assigned lecture section for the date and time.

ASYNCHRONOUS DESIGN STUDIO PREP represents one hour of unscheduled learning time per week (above and beyond the usual asynchronous time you would expect to spend reviewing lectures, studying, and doing assignments in a course). The activities are posted on Avenue to Learn prior to the start of each week to help you prepare for the Design Studio. The **asynchronous work for a given week must be completed before the Design Studio for that same week.**

DESIGN STUDIOS are a dedicated time for individual collaboration during the Design Studio's Individual weeks (3, 4, 5 and 8), team collaboration during the Design Studio's Group weeks (2, 9, 10, 11, and 12) and both during Design Studio weeks (6 and 7), on engineering economic project activities. Instructional staff (i.e. IAs and TAs) will be available for support during your scheduled time.

Online Management – Avenue-to-Learn

Avenue-to-Learn will be the online management system. Through Avenue, you will be able to:

- Find all course materials (lecture slides, project briefs, asynchronous tutorial worksheets and videos, design studio documents, etc.)
- View course-related announcements
- Submit coursework (individual quizzes and assignments and group work) for grading
- View your gradebook

6. Informal Course Description

The course lectures will deliver information to help you build the skills to apply engineering economics in decision-making. The Design Studios will be led and support the students in completing their engineering economic analysis and designs.

The topics in this course will be taught in sections divided among lecture weeks:

- Week 1: Course Introduction, Introduction to Engineering Economics, Project Introduction and Cost Estimation
- Week 2: Cost Benefit Analysis, Conversion Factors, and Technical Design
- Week 3: Project Management, Life Cycle Assessment, and Interest Rates
- Week 4: Inflation, Present Worth and Cash Flow Diagrams
- Week 5: Annual Worth, IRR and MARR
- Week 6: IRR Continued, Depreciation and Taxes
- Week 7: Reading Week
- Week 8: Risk Management and Sensitivity Analysis
- Week 9: Break Even Analysis, Scenario and Delay Analysis
- Week 10: Sensitivity Risk Management
- Week 11: Behavioral Economics, Supply and Demand
- Week 12: Exam Review
- Week 13: Exam Review
- Week 14: Field Trip – Capstone Expo

Design Studios consist of two case studies in Weeks 5 and 8 that include:

1. Consider a real-world example of given job opportunities
 - Applying topics of Conversion Factors, Cash Flow Analysis and PW calculations
2. Consider a real-world example of trade-off analysis in the context of depreciating assets
 - Applying topics of Depreciation and PW calculations

Along with a comprehensive project that includes:

1. Collaborating in multidisciplinary teams to develop a proposal for a large-scale infrastructure project.
 - Applying topics of Decision Making, Project Management, and Design
2. Assessing economic viability through cost estimation, cash flow analysis, and comparison methods.
 - Applying concepts of Time Value of Money, Inflation, and Comparison Methods to inform financial justification.
3. Perform sensitivity analysis to inform project decision-making

- Applying topics of Sensitivity Analysis and Monte Carlo Simulations

7. Student Deliverables and Assessment

The following is the grade breakdown for course assessments.

- Take-Home Quizzes (10%)
 - o 11 take-home quizzes; lowest quiz dropped.
- Case Studies (8%)*
 - o 2 Take-home case studies.
- DS Check-Ins (8%)
- DS Deliverables (6%)
 - o 3 DS deliverables (Weeks 3, 4, 7)
- Final Report (25%)
- Final Interview (5%)
- Main Project Self-Reflection (3%)
- Final Exam (35%)

* 4% each, subject to attendance regarding the scheme described below.

Take-Home Quizzes

A take-home quiz will be released every Tuesday 9:30 a.m. and will be due the following Tuesday at 8:09 a.m. for the duration of the term.

Case Studies

Students will individually consider scenarios where they are provided with different options, and each student will determine which option to pursue using engineering economics concepts such as PW calculations, conversion factors, cash flow diagrams and depreciation.

Main Project

Student teams will take on a real-world design problem that is provided to them, along with the project descriptions that detail the expectations. To select a project, students will need to identify where they will individually plan to use info from one or more of their past technical courses to "design" some aspect of the solution (i.e., use technical knowledge & analysis skills to come up with some relationship between cost and performance for some aspect of the project).

Note: you do not need to actually implement your designs by building anything or even design all aspects of the solution to work on a given project. You just need to design some aspects of it that would enable you to complete an economic analysis as detailed below.

Students will need to:

1. As a group, translate the project requirements into a formal economic assessment to determine feasibility and support decision-making.
2. Individually, "design" a subsystem of the facility (e.g., energy supply, transportation access, digital infrastructure, water/waste systems).

3. As a subgroup, combine results from individual design work to determine the most effective overall solution for the facility.
4. Present project progress through Check-Ins, final report, and the team interview.

The individual technical work students do is not rigorously assessed; rather, as long as your work has clearly been done with due diligence on your part, we will assess your presentation of the technical work and how well you use it to complete this economic analysis.

Group Work Policy

This course involves a lot of group work, and we encourage you to seek out strategies for conflict resolution and ensuring accountability amongst group members. Ideally, any difficulties should be addressed openly, and you should work together to find solutions.

If it is brought to our attention that a team member is not contributing fairly to a project, we will take appropriate action, which may include meeting with the students and, in some cases, adjusting individual grades.

To assess contributions fairly, we will accept evidence such as shared links to your collaborative online documents, which will allow us to track individual participation in different sections of the project. We strongly encourage students to communicate over Teams or Email for simpler contribution tracking. Additionally, we will monitor attendance and participation in Design Studio sessions. If it becomes clear that a team member has failed to make a fair contribution, we will adjust their grade accordingly. In extreme cases, this could result in a grade of zero for that student based on their lack of involvement.

Attendance Reweight

Attendance will be taken during both the lecture and Design Studio using the Top Hat app. Students should sign up for Top Hat using their McMaster emails to ensure that they are able to access course materials. For students who maintain an attendance rate of 80% or higher for both lectures and Design Studios, the weights of the assessments will be shifted to benefit students in the following way:

- The lower case study mark gets dropped, and the weight is shifted to the higher case study.

Students may review their attendance rate at any time by accessing the *Gradebook* section in the Top Hat application. This section provides a record of all attendance submissions along with the total number of attendance requests issued during the course.

8. Submission of Work

SUBMISSION PENALTIES: Please be aware of the following penalties for Individual Activities and Projects:

- All written reports and assignments must be uploaded to Avenue by 4:30 pm EST the day after your Design Studio, or they will be subject to a late penalty.
- Submissions must be in the correct format or will be subject to a mark deduction.
- Any submissions deemed to be partially or fully copied will be considered an academic offence and be subject to the terms laid out under the Academic Integrity Policy.

GRADING CONCERNS: Grades will be posted to Avenue as soon as possible upon completion.

Any questions/concerns must be addressed electronically through an MS Form that will be made available on Avenue.

- Concerns regarding grading will not be considered without submitting through the appropriate channels (i.e., the posted Microsoft Marking Concern Form): [ENGINEER 3PX3 Marking Concerns Form – Fill out form](#)

9. Important Dates

Case Study 1	Design Studio Week 5
Case Study 2	Design Studio Week 8
Final Report Due	Design Studio Week 11
Final Team Interview	Design Studio Week 12
Main Project Self-Reflection Due	Design Studio Week 12

Dates are subject to change. Appropriate notice will be provided. Refer to Lecture 01 – Introduction and Project Modules for more.

10. Textbooks and Course Materials

No required textbooks. All required course materials will be provided on Avenue to Learn.

11. Policy Statements

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at [University Secretariat - McMaster University Policy Directory](#).

Generative AI: Some Use Permitted

Students may use generative AI in this course so long as the use of generative AI is referenced and cited following the citation instructions given in the course. Students are not permitted to use generative AI to produce original work for any assessment. All prompts and outputs must be included in the appendix of the submission, along with a description of how the tool was used and must be cited appropriately. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the

student's responsibility to be clear on the expectations for citation and reference and to do so appropriately. Please refer to [Guidelines on the Use of Generative AI in Teaching and Learning](#) for further information on this policy.

The following illustrates only four forms of academic dishonesty:

- Plagiarism, e.g. submission of work not one's own or which other credit been obtained.
- Improper collaboration in group work.
- Copying or using unauthorized aids in tests and examinations.
- Using generative AI to complete quizzes or assist with report writing.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. In those courses, students will be expected to submit their work electronically either directly to Turnitin.com or via Avenue to Learn (A2L) plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish to submit their work through A2L and/or Turnitin.com must still submit an electronic and/or hardcopy to the instructor. No penalty will be assigned to a student who does not submit work to Turnitin.com or A2L. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). To see the Turnitin.com Policy, please go to www.mcmaster.ca/academicintegrity.

ON-LINE ACCESS

This course will use **Avenue-to-Learn** and **Microsoft Teams** for communication. Students should be aware that, when they access the electronic components of this course, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university

functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services \(SAS\)](#) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

It's the student's responsibility to activate their accommodations at their earliest convenience and before due dates for assignments. Accommodations cannot be made retroactively for due dates that have passed prior to the activation of an accommodation plan.

ACADEMIC ACCOMMODATION FOR RELIEF FOR MISSED ACADEMIC WORK – McMaster Student Absence Form (MSAF)

In the event of an absence for medical or other reasons, students should review and follow the "Requests for Relief for Missed Academic Term Work".

1. All MSAFs are to be directed to engic@mcmaster.ca, not to a professor in this course. Sending to another email address will delay processing.
2. If you do not contact engic@mcmaster.ca after submitting your MSAF, your MSAF may not be processed.
3. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in their course.
 - For **individual assignments**, the assessment deadline will be extended by **4 days**. This includes the Technical Design Deliverable, Project Management and CBA Deliverable, Case Study 1, Case Study 2, and the Main Project Self-Reflection.
 - MSAFs for the project **interview** will be rescheduled to a later date.
 - **Team or pair deliverables CAN NOT** have an MSAF applied to them. This includes the Comparison Methods 2 Check-in and the Final Report.

The following table represents a Wednesday Design Studio section; if your DS section is on Monday or Tuesday, the dates will vary.

Assignment	Type	Due Date/Occurrence	SAS Accommodation	MSAF Action	Latest Acceptable Submission
Cost Estimation and Financial Model Development	Individual	Thu., Jan. 22, 2026 4:30 PM	4 Day Extension	4 Day Extension	Mon., Jan. 26, 2026 4:30 PM

Project Management	Individual	Thu., Jan. 29, 2026 4:30 PM	4 Day Extension	4 Day Extension	Mon., Feb. 02, 2026 4:30 PM
Case Study 1	Individual	Thu., Feb. 05, 2026 4:30 PM	4 Day Extension	4 Day Extension	Mon., Feb. 09, 2026 4:30 PM
Economic Refinement and Robustness Analysis	Pairs	Thu., Feb. 26, 2026 4:30 PM	Unapplicable	Unapplicable	
Case Study 2	Individual	Thu., Mar. 05, 2026 4:30 PM	4 Day Extension	4 Day Extension	Mon., Mar. 09, 2026 4:30 PM
Final Report	Group	Thu., Mar. 26, 2026 4:30 PM	Unapplicable	Unapplicable	
Interviews	Individual	Mon., Mar. 30, 2026	1 Week Reschedule	1 Week Reschedule	Mon., Apr. 06, 2026
Main Project Self-Reflection	Individual	Fri., Apr. 10, 2026 4:30 PM	4 Day Extension	4 Day Extension	Tue., Apr. 14, 2026 4:30 PM

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor or any authorized personnel for authorized distribution, or by a student for personal study. Students should be aware that their voices and/or images may be recorded by others during the class, tutorial, or Design Studio. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

NOTICE REGARDING POSSIBLE COURSE MODIFICATION

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

REFERENCE TO RESEARCH ETHICS

The two principles underlying integrity in research in a university setting are these: a researcher must be honest in proposing, seeking support for, conducting, and reporting research; a researcher must respect the rights of others in these activities. Any departure from these principles will diminish the integrity of the research enterprise. This policy applies to all those conducting research at or under the aegis of McMaster University. It is incumbent upon all members of the university community to practice and to promote ethical behaviour. To see the Policy on Research Ethics at McMaster University, please go to <https://reo.mcmaster.ca/>.